WORK SHEET - 06 **STRAIGHT LINE AND CIRCLE**

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.12 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 Let P and Q be any two points on the circle $x^2+y^2=4$ such that PQ is a diameter. If L_1 and L_2 are the lengths of perpendiculars from P and Q on $x+y=1$, then the maximum value of L_1L_2 is

(A) 1/2 (B) 7/2 (C) 1 (D) 2

Q.2 Let $A(1, 2)$, $B(3, 4)$ and $C(x, y)$ be such that $(x-1)(x-3)+(y-2)(y-4)=0$. If area of $\Delta ABC = 1$, then number of positions of C in xy plane, is

(A) 2 (B) 4 (C) 8 (D) 16

Q.3 Line $x+y=2$ is common tangent of two circles C_1 and C_2 with radii 1 and 2 and its points of contact are $A(1, 1)$ and B respectively. Line $2x-y=4$ is the radical axis of C_1 and C_2, then B will be

(A) (4, –2) (B) (3, –1) (C) $\left(\frac{5}{2}, \frac{-1}{2}\right)$ (D) (2, 0)

Q.4 Let $A(3, 4)$, $B(0, 0)$ and $C(3, 0)$ be vertices of ΔABC. If P is a point inside ΔABC, such that $d(P, BC) \le \min\left(d(P, AB), d(P, AC)\right)$, then the maximum value of $d(P, BC)$ is equal to

[**Note :** d(P, BC) represents distance of P from BC.]

(A) 1 (B) 1/2 (C) 2 (D) 3/4

Q.5 In triangle ABC, the side lengths BC, CA and AB are consecutive positive integers in increasing order. Let the equations $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$ represent the lines AB and AC respectively and $\left|\frac{a_1b_2-a_2b_1}{a_1a_2+b_1b_2}\right| = \frac{4}{3}$, then the value of $\left(\frac{BC+CA-AB}{2}\right)$ is equal to

(A) 2 (B) 4 (C) 6 (D) 12

Q.6 Consider a family of circles given by $C_n : \left(x-\frac{1}{n(n+1)}\right)^2 + y^2 = \beta_n^{\,2}, n \in N$ such that each member of the family touches the line pair $x^2-y^2=0$ then value of $\sum_{n=2}^{\infty}\beta_n$ equals

(A) $\frac{1}{2\sqrt{2}}$ (B) $\sqrt{2}$ (C) $\frac{1}{\sqrt{2}}$ (D) 2

Q.7 If $A \equiv (3, 1+\sin\theta)$ and $B \equiv (-1, 1+\cos\theta)$, $\theta \in R$ are two points and P is a point on the line $x-y=0$ then minimum value of $(PA+PB)$ is

(A) 4 (B) $4+2\sqrt{2}$ (C) $4-2\sqrt{2}$ (D) 2

Q.8 Let $A = \{(x_1, y_1) \mid 3x_1-4y_1-3=0\}$ and $B = \{(x_2, y_2) \mid (x_2-3)^2+(y_2-4)^2=0\}$

If $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$ is minimum, then $(x_1, y_1) \equiv (a, b)$, where $a-b=\frac{p}{q}$, $p, q \in N$, the least value of $(p-q)$, is

(A) 3 (B) 2 (C) 4 (D) 1

Q.9 The coordinates of four vertices of a quadrilateral are $A(-2, 4)$, $B(-1, 2)$, $C(1, 2)$ and $D(2, 4)$ taken in order. The equation of line passing through the vertex $(-1, 2)$ and dividing the quadrilateral in two equal areas is

(A) $x+1=0$ (B) $y=2$ (C) $x+y=1$ (D) $x-y+3=0$

Q.10 If lines L_1 and L_2 are two tangents to a circle $x^2 + y^2 = 4$ and $y = \sqrt{20}$ is one of the angle bisector between L_1 and L_2 then product of their slopes is
(A) 8 (B) 4 (C) – 4 (D) – 8

Q.11 ABC is an isosceles triangle in which A is (–1, 0), $\angle A = \frac{2\pi}{3}$, AB = AC and AB is along the x-axis. If $BC = 4\sqrt{3}$ then the equation of line BC is
(A) $x + \sqrt{3}\,y = 3$ (B) $\sqrt{3}\,x + y = 3$ (C) $x - \sqrt{3}\,y + 5 = 0$ (D) $3x + y = \sqrt{3}$

Q.12 Let P(1, 1) and Q(3, 2) then the point R on x-axis such that PR + RQ is minimum, is
(A) $\left(\frac{5}{3}, 0\right)$ (B) $\left(\frac{2}{3}, 0\right)$ (C) $\left(\frac{1}{3}, 0\right)$ (D) (3, 0)

[PARAGRAPH TYPE]

Q.13 to Q.15 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 13 to 15

Let L be the line belonging to the family of straight lines $(m + 2n)x + (m - 3n)y + (m - 8n) = 0$, $m, n \in R$, which is farthest from the point P(2, 2).

Q.13 The equation of line L, is
(A) $x + 4y + 7 = 0$ (B) $2x + 3y + 4 = 0$ (C) $4x - y - 6 = 0$ (D) $x + y + 1 = 0$

Q.14 Area of triangle formed by line L with coordinate axes is
(A) $\frac{1}{2}$ (B) $\frac{4}{3}$ (C) $\frac{49}{8}$ (D) $\frac{9}{2}$

Q.15 If line L is concurrent with the lines $x - 2y + 1 = 0$ and $3x - 4y + c = 0$ then the value of c is
(A) – 5 (B) 5 (C) – 4 (D) 4

[MULTIPLE CORRECT CHOICE TYPE]

Q.16 to Q.18 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

Q.16 A circle touches the line $x - y = 0$ at a point P such that $OP = 4\sqrt{2}$ where O is origin. The circle contains the point (– 10, 2) in its interior and the length of its chord-intercept on the line $x + y = 0$ is $6\sqrt{2}$. If the circle equation is written in the form $x^2 + y^2 + ax + by + c = 0$, then $(a + b + c)$ is less than
(A) 40 (B) 48 (C) 56 (D) 64

Q.17 The equations of two equal sides AB and AC of an isosceles triangle ABC are $x + y = 5$ and $7x - y = 3$ respectively. Then the equations of sides BC, if area $(\Delta ABC) = 5$, is
(A) $x - 3y + 1 = 0$ (B) $x - 3y - 21 = 0$ (C) $3x + y + 2 = 0$ (D) $3x + y - 12 = 0$

Q.18 Perpendicular is drawn from a fixed point (3, 4) to a variable line having x-intercept unity. $P(x, y) = 0$ represents the locus of the foot of perpendicular drawn from (3, 4) to the variable line, which is a circle. Which of the following statement(s) is(are) correct?
(A) Radius of circle is $\sqrt{3}$.
(B) Radius of circle $\sqrt{5}$.
(C) If tangent is drawn to $P(x, y) = 0$ from the origin, then the length of tangent is $\sqrt{3}$.
(D) If tangent is drawn to $P(x, y) = 0$ from the origin then the length of tangent is $\sqrt{5}$.

[INTEGER TYPE]

Q.19 & Q.20 are "Integer Type" questions. (The answer to each of the questions **are upto 4 digits**)

Q.19 Let S_1 and S_2 are the unit circles with centre at $C_1(0, 0)$ and $C_2(1, 0)$ respectively. Let S_3 is another circle of unit radius, passes through C_1 and C_2 and its centre is above the x-axis. If equation of common tangent to S_1 and S_3, which does not pass through S_2, is $ax + by + 2 = 0$, then find the value of $(a^2 - b)$.

Q.20 If the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ cuts each of the circles
$x^2 + y^2 - 4 = 0$, $x^2 + y^2 - 6x - 8y + 10 = 0$ and $x^2 + y^2 + 2x - 4y - 2 = 0$
at the extremities of diameter, then find the value of $(g^2 + f^2 - c)$.

ANSWER KEY

Q.1	B	Q.2	B	Q.3	B	Q.4	A	Q.5	C
Q.6	A	Q.7	A	Q.8	C	Q.9	D	Q.10	C
Q.11	AC	Q.12	A	Q.13	A	Q.14	C	Q.15	B
Q.16	CD	Q.17	AD	Q.18	BC	Q.19	[4]	Q.20	[17]